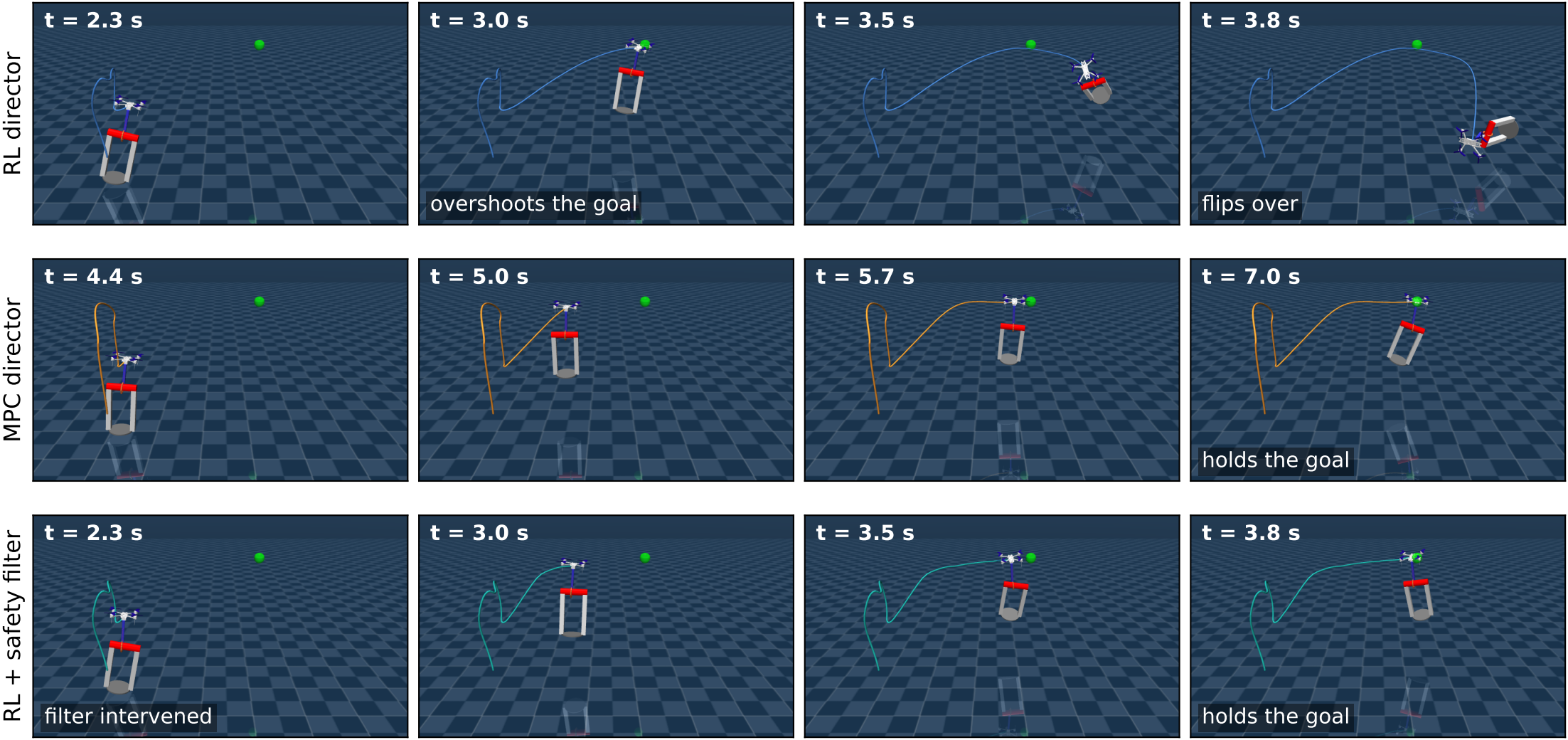

RL director
t = 2.3 s
t = 3.0 s
overshoots the goal
t = 3.5 s
t = 3.8 s
flips over
MPC director
t = 4.4 s
t = 5.0 s
t = 5.7 s
t = 7.0 s
holds the goal
RL + safety filter
t = 2.3 s
filter intervened
t = 3.0 s
t = 3.5 s
t = 3.8 s
holds the goal